

Limitations

Ethereum applications have their own downsides. These decentralised applications aren't faultless. Since smart contract codes are written by humans, code bugs or oversights may occur. This may lead to unintended adverse actions. If a mistake in the code gets exploited, there is no efficient way to stop this attack or exploitation. Only obtaining network consensus and rewriting the underlying code can solve such problems. This challenges the supposed immutability of the Ethereum platform.

Further, Ethereum has too many platforms. This platform has a ledger, a smart contracts generator and many more platforms that make it weak. Also, it is not possible to upgrade the Ethereum system so easily.

Programming languages used in Ethereum

Ethereum mainly depends upon the smart contract system. Smart contracts are essentially programs that exist on the Ethereum blockchain. They perform various tasks such as sending or receiving Ether or ERC 20 tokens, among other things. Solidity and Serpent are some of the most popular programming languages used to write smart contracts in Ethereum.

Gradually, Ethereum is becoming a sustainable cryptocurrency. There is long term scope for Ethereum and that is why its price is rising, just like Bitcoin.

The Ethereum network consists of over 27,500 nodes and each node runs EVM implementation. C++, Python, Ruby, Go, JavaScript, Rust, Haskell and many more languages are used to program the Ethereum network. Many companies like Microsoft, IBM, Deloitte, Barclays, Credit Suisse and UBS are using dApps or decentralised applications of Ethereum.

The most popular application of Ethereum is its crowdfunding platform. Other than that, it is widely

used in digital signature algorithms and securitised tokens. It is also suitable for digital rights management, online gambling, financial applications, systems identification and much more.

The programming languages mentioned earlier help the Ethereum network to:

- Mine real Ether
- Transfer funds between addresses
- Create contracts and send transactions
- Explore block history, etc.

Layers of Ethereum

The logic and mechanisms behind Ethereum work on the layers it employs. These layers include the following.

The Ethereum hardware layer:

The Ethereum hardware layer is a peer-to-peer network of computers that compute transactions and keep them in order in a shared ledger. Transactions can contain both values and information. The value is expressed in units called Ethers — the digital currency of the Ethereum platform and the information is code that can pass data and trigger actions. In the Ethereum hardware layer, there are hundreds of computers spread across the planet that are connected to one another via the Internet.

The Ethereum software layer: The Ethereum software layer has been built to allow the transaction of values in any shape or form. This may be currency, a house, an identity, the rights to use or reproduce a song, or any other asset you

think of. Thus Ethereum's capabilities extend far beyond that of Bitcoin, beyond just currency. All Ethereum code is open source and available on *Github.com*.

The Ethereum application

layer — dApps: The combination of hardware and software layers described above has made Ethereum work as a global, decentralised computer on which third party applications can be developed.

There are more than 900 applications built on Ethereum. About 30 per cent of them are live while the other 70 per cent are still under development. Building these public, decentralised applications has several benefits like transparency, resilience, better code, etc.

The road ahead

Until now, enterprises have dealt with the blockchain in numerous different ways and with a variety of infrastructure options. But the blockchain platform applied to Ethereum is said to move decisively towards better and more secure cryptocurrency.

Ethereum presently is facing the problem of scalability. Several projects like Plasma, Sharding and Raiden are working to solve this problem. According to the Investing Haven, Ethereum's price will reach US\$ 1,000 by the year 2020. And it is the immutability and flexibility of the platform that are likely to act in favour of Ethereum. **END** 

References

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